MSD MANUAL
Consumer Version

Chronic Kidney Disease

(Chronic Kidney Failure; Chronic Renal Failure; CKD)

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Chronic kidney disease is a slowly progressive (months to years) decline in the kidneys' ability to filter metabolic waste products from the blood.

- Major causes are diabetes and high blood pressure.
- Blood becomes more acidic, anemia develops, nerves are damaged, bone tissue deteriorates, and risk of atherosclerosis increases.
- Symptoms can include urinating at night, fatigue, nausea, itching, muscle twitching and cramps, loss of appetite, confusion, difficulty breathing, and body swelling (most commonly the legs).
- Diagnosis is by blood and urine tests.
- Treatment aims to restrict fluids, sodium, and potassium in the diet; use medications to correct other conditions (such as diabetes, high blood pressure, anemia, and electrolyte imbalances); and, when necessary, use dialysis or kidney transplantation.

(See also Overview of Kidney Failure.)

Many diseases can irreversibly damage or injure the kidneys. Acute kidney injury becomes chronic kidney disease if kidney function does not recover after treatment and lasts more than 3 months. Therefore, anything that can cause acute kidney injury can cause chronic kidney disease. However, in Western countries, the **most common causes** of chronic kidney disease are

- Diabetes mellitus
- High blood pressure (hypertension)

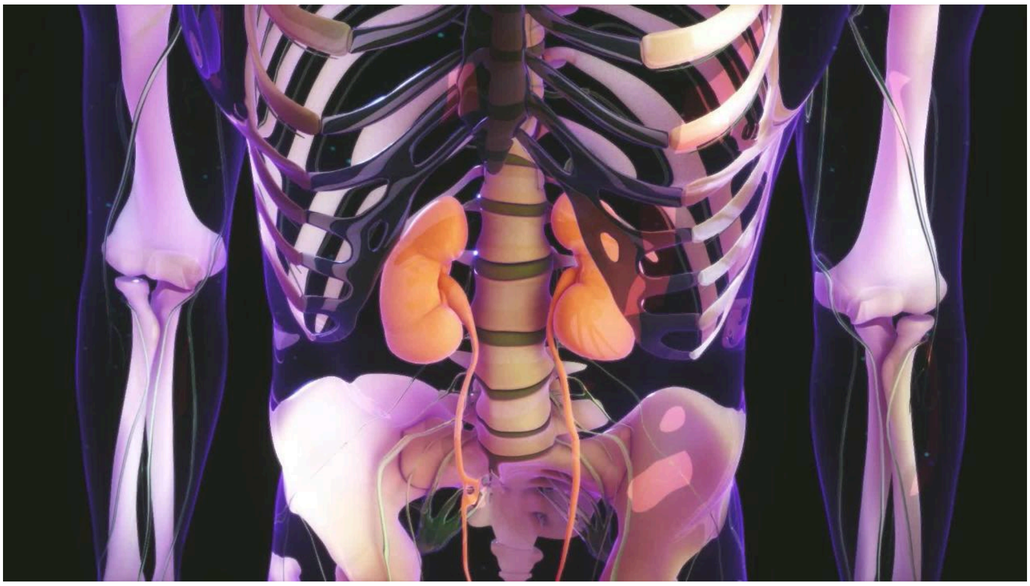
Both of these conditions directly damage the kidneys' small blood vessels.

Other causes of chronic kidney disease include urinary tract blockage (obstruction), certain kidney abnormalities (such as polycystic kidney disease and glomerulonephritis), and autoimmune disorders (such as systemic lupus erythematosus [lupus]) in which antibodies damage the tiny blood vessels (glomeruli) and the tiny tubes (tubules) of the kidneys.

Chronic kidney disease causes many problems throughout the body:

The Kidneys

VIDEO



- When loss of kidney function is mild or moderately severe, the kidneys cannot absorb water from the urine to reduce the volume of urine and concentrate it.
- Later, the kidneys have less ability to excrete the acids normally produced by the body and the blood becomes more acidic, a condition called acidosis.
- The ability to excrete potassium decreases, leading to high levels in the blood, a condition called hyperkalemia.
- Production of red blood cells decreases, leading to anemia.
- High levels of metabolic waste products in the blood can damage nerve cells in the brain, trunk, arms, and legs. Uric acid levels may increase, sometimes causing gout.
- Diseased kidneys cannot excrete excess salt and water. Salt and water retention can contribute to high blood pressure and heart failure.
- The sac that surrounds the heart (pericardium) may become inflamed (pericarditis).
- The level of triglycerides in the blood is often elevated, which, along with high blood pressure, increases the risk of atherosclerosis.
- The formation and maintenance of bone tissue may be impaired (renal osteodystrophy) if certain conditions that accompany chronic kidney disease are present for a long time. These conditions include a high level of parathyroid hormone, a low concentration of calcitriol (the active form of vitamin D) in the blood, impaired absorption of calcium, and a high concentration of phosphate in the blood. Renal osteodystrophy may lead to bone pain and an increased risk of fractures.

Symptoms of Chronic Kidney Disease

Symptoms usually develop very slowly. As kidney failure progresses and metabolic waste products build up in the blood, symptoms progress.

Mild to moderate loss of kidney function may cause only mild symptoms, such as the need to urinate several times during the night (nocturia). Nocturia occurs because the kidneys cannot absorb water from the urine to reduce the volume and concentrate it as normally occurs during the night.

As kidney function worsens and more metabolic waste products build up in the blood, people may feel fatigued and generally weak and may become less mentally alert. Some have a loss of appetite and shortness of breath. Anemia also contributes to fatigue and generalized weakness.

The buildup of metabolic waste also causes loss of appetite, nausea, vomiting, and an unpleasant taste in the mouth, which may lead to undernutrition and weight loss. People with chronic kidney disease tend to bruise easily or bleed for an unusually long time after cuts or other injuries. Chronic kidney disease also diminishes the body's ability to fight infections. Gout may cause acute arthritis with joint pain and swelling.

Severe loss of kidney function causes metabolic wastes to build up to higher levels in the blood. Damage to muscles and nerves can cause muscle twitches, muscle weakness, cramps, and pain. People may also feel a pins-and-needles sensation in the arms and legs and may lose sensation in certain areas of the body. They may develop restless legs syndrome. Encephalopathy, a condition in which the brain malfunctions, may ensue and lead to confusion, lethargy, and seizures.

Heart failure may cause shortness of breath. Body swelling may develop, particularly in the legs. Pericarditis may cause chest pain and low blood pressure. People who have advanced chronic kidney disease commonly develop gastrointestinal ulcers and bleeding. The skin may turn yellow-brown and/or dry, and occasionally, the concentration of urea is so high that it crystallizes from sweat, forming a white powder on the skin. Some people with chronic kidney disease itch all over their body. Their breath may also be foul.

Diagnosis of Chronic Kidney Disease

- Blood and urine tests
- Ultrasound
- Sometimes biopsy

Blood and urine tests are essential. They confirm the decline in kidney function.

When loss of kidney function reaches a certain level in chronic kidney disease, the levels of chemicals in the blood typically become abnormal.

- Urea and creatinine, metabolic waste products that are normally filtered out by the kidneys, are increased.
- Blood becomes moderately acidic.

- Potassium in the blood is often normal or only slightly increased but can become dangerously high.
- Calcium and calcitriol in the blood decrease.
- Phosphate and parathyroid hormone levels increase.
- Hemoglobin is usually lower (which means the person has some degree of anemia).

Doctors measure kidney function by using the level of creatinine in the blood, sex, and body weight in a formula called the estimated glomerular filtration rate (eGFR). In the past, some of these formulas used race to assess whether kidney function was abnormal. However, doing so increased health care disparities between races in diagnosis and treatment of kidney disease. Thus, including race in such assessments is no longer recommended.

Measuring the blood level of potassium is important, because it can become dangerously high when kidney failure reaches an advanced stage or if people ingest large amounts of potassium or take a medication that prevents the kidneys from excreting the potassium.

Analysis of the urine may detect many abnormalities, including protein and abnormal cells.

Ultrasonography is often done to rule out obstruction and check the size of the kidneys. Small, scarred kidneys often indicate that loss of kidney function is chronic. Determining a precise cause becomes increasingly difficult as chronic kidney disease reaches an advanced stage.

Removing a sample of tissue from a kidney for examination (kidney biopsy) may be the most accurate test, but it is not recommended if results of an ultrasound examination show that the kidneys are already small and scarred.

Treatment of Chronic Kidney Disease

- Treatment of conditions that worsen kidney function
- Dietary measures and medications
- Dialysis or kidney transplantation

The goal of treatment is to slow the decline of kidney function and delay the need for dialysis.

Conditions that can cause or worsen chronic kidney disease and adversely affect overall health should be promptly addressed, such as

- Diabetes
- High blood pressure (hypertension)
- Urinary tract obstruction
- Infections
- Use of certain medications

Controlling the level of sugar (glucose) in the blood as well as high blood pressure in people with diabetes substantially slows deterioration in kidney function. Medications called angiotensin-converting enzyme (ACE) inhibitors and angiotensin II receptor blockers (ARBs), which help lower blood pressure, may decrease the rate of decline in kidney function in some people with chronic kidney disease. Medications called sodium-glucose cotransporter-2 (SGLT2) inhibitors can also slow deterioration in kidney function, but they should be avoided in people with type 1 diabetes mellitus.

Doctors avoid prescribing medications that are excreted by the kidneys, or they prescribe lower doses of such medications. Many other medications may need to be avoided. For example, ACE inhibitors, ARBs, and certain diuretics (such as spironolactone, amiloride, and triamterene) may need to be stopped in people with severe chronic kidney disease and high potassium levels because these medications can increase potassium levels.

Obstructions in the urinary tract are removed or relieved. Bacterial infections are treated with antibiotics.

Dietary measures should be taken.

Restricting protein

The decline in kidney function can be slowed slightly by restricting the amount of protein consumed daily. People need to consume sufficient carbohydrates to offset the reduction in protein. If dietary protein is significantly restricted, it is wise to have the supervision of a dietitian to be sure adequate amounts of amino acids are taken in.

Controlling acidosis

Sometimes, mild acidosis can be controlled by increasing the intake of fruits and vegetables and reducing the intake of animal protein. However, moderate or severe acidosis may require treatment with acid-lowering medications (for example, sodium bicarbonate and sodium citrate).

Lowering triglyceride levels

The triglyceride and cholesterol levels in the blood may be controlled somewhat by limiting fat in the diet. Medications such as statins, ezetimibe, or both may be required to lower the levels of triglycerides and cholesterol.

Restricting sodium and potassium

The restriction of salt (sodium) is usually beneficial, especially if the person has heart failure.

Fluid intake may need to be restricted to prevent the sodium concentration in the blood from becoming too low. Foods that are extremely high in potassium, such as salt substitutes, must be avoided, and foods that are somewhat high in potassium, such as dates, figs, and many other fruits, should not be consumed in excess. (See the National Kidney Foundation’s publication Potassium in Your CKD Diet for more information.)

A high potassium level in the blood increases the risk of abnormal heart rhythms and cardiac arrest. If the potassium level becomes too high, potassium-lowering medications (for example, sodium polystyrene sulfonate, patiromer, and zirconium cyclosilicate) may help, but emergency dialysis may be required.

Controlling phosphorus levels

The elevated phosphorus level in the blood can cause deposits of calcium and phosphorus to form in tissues, including the blood vessels. Restricting the intake of foods that are high in phosphorus, such as dairy products, liver, legumes, nuts, and most soft drinks, lowers the phosphate concentration in the blood. Medications that bind phosphate, such as calcium carbonate, calcium acetate, sevelamer, lanthanum, and ferric citrate, taken by mouth, may also lower the phosphorus level in the blood. Calcium citrate should be avoided. Calcium citrate is found in many calcium supplements and is in many products as a food additive (sometimes called E333). Vitamin D and similar medications are often taken by mouth to reduce high levels of parathyroid hormone.

Treating complications

The **anemia** caused by chronic kidney disease is treated with

- Medications such as erythropoietin or darbepoietin
- Blood transfusions

Doctors also look for and treat other causes of anemia, particularly dietary deficiencies of iron, folate (folic acid), and vitamin B12 (see Vitamin Deficiency Anemia); and gastrointestinal bleeding (blood loss through the digestive tract).

Most people who take erythropoietin or darbepoietin regularly need to be given iron intravenously to prevent iron deficiency, which impairs the body’s response to these medications. Erythropoietin and darbepoietin should be used only when necessary because they can increase the risk of stroke. The tendency to bleed can be temporarily suppressed by transfusions of blood products or by such medications as desmopressin or estrogens. Such treatment may be needed after an injury or before a surgical procedure or a tooth extraction.

Blood transfusions are given only if the anemia is severe, is causing symptoms, and does not respond to erythropoietin or darbepoietin.

High blood pressure is treated with antihypertensive medications to prevent further impairment of heart and kidney function.

Diuretics may also relieve symptoms of **heart failure**, even when kidney function is poor, but dialysis may be needed to remove the excess body water in severe chronic kidney disease.

Treating advanced chronic kidney disease

When the treatments for chronic kidney disease are no longer effective, the only options are long-term dialysis and kidney transplantation. Both options decrease symptoms and prolong life. If the person is a candidate, kidney transplantation can be an excellent option. For people who choose not to undergo dialysis, end-of-life care (also called hospice, a kind of palliative care) is important.

Prognosis for Chronic Kidney Disease

If chronic kidney disease is caused by a disorder that can be corrected (for example, urinary tract blockage) and if that disorder has not been present for too long, then kidney function may improve when the causative disorder is successfully treated. Otherwise, kidney function tends to worsen over time. The rate of decline in kidney function depends somewhat on the underlying disorder causing chronic kidney disease and on how well the disorder is controlled. For example, diabetes and high blood pressure, particularly if poorly controlled, cause kidney function to decline more rapidly. Chronic kidney disease is fatal if not treated.

When the decline in kidney function is severe (sometimes called end-stage kidney failure or end-stage kidney disease), survival is usually limited to several months in people who are not treated, but people who are treated with dialysis can live much longer. However, even with dialysis, people with end-stage kidney failure die sooner than people their age who do not have end-stage kidney disease. Most die from heart or blood vessel disorders or infections.

More Information

The following English-language resources may be useful. Please note that The Manual is not responsible for the content of these resources.

- National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)
- National Kidney Foundation (NKF)